

HawkEye Traffic AI

Contextualizing Computer Vision for Urban Traffic Enforcement

Concept Note

Gridlock Hackathon 2.0

Flipkart Prototype

Prepared for: Flipkart Gridlock Hackathon Evaluation Panel

Project Scope: Edge-Ready Asynchronous Traffic Violation Detection System

1. The Core Problem: The Context Gap in Legacy AI

Traditional computer vision pipelines deployed for traffic enforcement face a massive constraint: they are completely devoid of context. A standard object detection model (like vanilla YOLO) evaluating a Bengaluru intersection will blindly issue a "Triple Riding" fine if it counts three passengers on a motorcycle, regardless of whether the third passenger is a 4-year-old child. Similarly, legacy systems analyze static frames, rendering them mathematically incapable of detecting complex vector violations like Wrong-Side Driving or Illegal Parking. This leads to severe false-positive rates, citizen outrage, and wasted administrative resources.

2. The Innovation: Smart Enforcement Validation Engine (SEVE)

HawkEye Traffic AI is a paradigm shift. We do not just detect objects; we evaluate their relationships. HawkEye introduces the **Smart Enforcement Validation Engine (SEVE)**, a secondary contextual logic layer that acts as an automated judge before any fine is issued.

- **Child Passenger Suppression:** SEVE calculates the bounding-box area ratio of all passengers on a detected motorcycle. If the smallest rider occupies less than 55% of the volumetric space of the largest rider, the engine classifies them as a child and mathematically overrides the Triple Riding flag.
- **Positional Logic:** Helmets are not just detected; their coordinates are cross-referenced with the rider's physical anatomy to ensure the helmet is worn on the head, not carried on the handlebar.
- **Environment Quality Score (EQS):** Before processing, SEVE measures Laplacian Variance and HSV Intensity to reject blurry or low-light photos entirely, ensuring only high-fidelity evidence reaches the OCR stage.

3. Temporal Memory: BoT-SORT Evidence Bursts

Instead of processing single static images, HawkEye processes "Evidence Bursts"—sequential image frames from a traffic camera. By integrating **BoT-SORT (Bag of Tricks for Simple Online Realtime Tracking)**, HawkEye establishes temporal memory. The AI extracts a trajectory vector (dx, dy) for every vehicle, enabling flawless detection of **Wrong-Side Driving** (motion opposing lane flow) and **Overspeeding**.

4. Precision OCR Pipeline

HawkEye eliminates the common issue of OCR models hallucinating text from billboards or clothing. The system utilizes YOLO to isolate and strictly crop the license plate coordinates, applies a 5-pixel expansion margin, converts the array to high-contrast Grayscale, and pushes this micro-crop to EasyOCR. This ensures absolute precision under adverse urban conditions.

"HawkEye does not replace traffic police with cameras; it replaces arbitrary algorithms with contextual intelligence, ensuring every issued challan is mathematically sound and contextually fair."

5. Enterprise-Ready Architecture

The solution is not a monolithic script; it is a scalable cloud-native application. HawkEye leverages **FastAPI** for rapid data ingestion and routes heavy ML workloads to asynchronous **Celery workers** via a **Redis** message broker. All validated evidence is committed to a **MongoDB** NoSQL database. Finally, a **Streamlit Command Center** serves as the live UI, allowing traffic officers to view automated evidence rendering and search for repeat offenders.

Conclusion

HawkEye Traffic AI directly satisfies and exceeds the Flipkart Gridlock Hackathon problem statement by providing a highly scalable, contextually aware, and edge-deployable prototype that solves the "False Positive" crisis in Indian traffic enforcement.